



Web-based 3D Visualization System for Anatomy Online Instruction

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ABSTRACT

Problem-based instruction is an active learning instructional practice that requires students to use rational and critical thinking skills to generate reasonable solutions to problem-based scenarios. For complex medical conditions such as stroke, degenerative diseases, and traumatic brain injury, students must have a strong command of neuroanatomy and physiology. While virtual and synthetic dissection simulation tools alleviate the need for procuring and maintaining costly resources, like cadavers, these tools are costly, inaccessible to students online, and inadequate in the teaching of practical knowledge needed to solve real-life clinical problems. In the wake of the Covid-19 pandemic, many courses have switched to an online format surprising students and faculty. The web-based visualization repository presented is intended to provide medical students with a comprehensive, web-based visual and problem-based learning tool to assist their learning of anatomical and neurophysiological concepts as applied to various medical disorders. The application can be used for online learning, as well as for in-person learning.

CCS CONCEPTS

• **Human-centered computing** → **Visualization systems and tools**.

KEYWORDS

Web 3D, Neuroanatomy, e-Learning, Problem-based Learning

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1 INTRODUCTION

For those entering the field of medical education (e.g., speech-language pathology, neurosurgery), the importance of a strong foundation in neuroanatomy and neurophysiology cannot be understated. Speech-language pathologists (SLPs) require an understanding of the brain and how it works in general, as well as comprehensive knowledge of the specific aspects of the brain that govern language, cognition, and sensorimotor control for speech production and swallowing. However, acquiring the requisite knowledge over a relatively short amount of time (e.g., a few semesters) can be difficult. Instructors often struggle to adequately convey this complex information in ways that serve all of their students, who have different levels of background knowledge and varied learning styles [15]. Historically, neuroanatomy and physiology instruction in speech-language pathology have relied on traditional, lecture-based instruction supplemented with hands-on training in gross anatomy laboratories [12]. Rising costs, resource demands, a global pandemic, and higher education's adoption of more active learning instructional practices have made these instructional practices impractical and outdated. While virtual simulation tools alleviate the need for procuring and maintaining costly resources, like cadavers, these tools require complex setups and are sometimes expensive to customize for teaching practical knowledge needed to solve real-life clinical scenarios.

Problem-based instruction is an active learning instructional practice that teaches students rational and critical thinking skills, in order to generate reasonable solutions to problem-based scenarios [7]. For example, in a problem-based learning environment, students may be required to diagnose a fictitious patient's communication or swallowing disorder based on specific symptoms. For complex medical conditions such as stroke, degenerative diseases, and traumatic brain injury, completing these tasks would require students to have a strong command of neuroanatomy and physiology, as well as other discipline-specific knowledge. Problem-based learning, in addition to lecture and gross anatomy lab-based instruction, facilitates knowledge transfer and improves the critical thinking skills required for clinical practice.

The realities of Covid-19 imposed physical barriers to learning and created an increased focus on ways to leverage technology to deliver high-quality remote instruction. Further complicating the situation, many students find themselves and their families in

difficult circumstances, particularly financially. Likewise, higher learning institutions are facing funding woes and uncertainty, hence most are not likely to spend resources on laboratory-based high-end anatomy and physiology instructional technology.

2 RELATED WORK

The use of 3D models for anatomical visualization is not new, as there are established proprietary hardware and software systems with 3D anatomical data sets. Many commercial products are available for 3D medical visualization, such as Vesalius3D [13] or the Anatomage Table [2]. Anatomage uses custom-built PC hardware platforms, a combination of horizontal infrared touch displays and wall-mounted monitors, along with a pre-installed software package with multiple 3D human models, several disease case-studies, as well as touch controls to perform virtual 3D dissections. Visualization of 3D human models and the extensive case study library paired with the interactive virtual format enables the time frame of virtual dissection labs to be extended beyond what would normally be possible in physical dissection labs. However, these software/hardware systems are not accessible to students outside the lab environment due to the static nature of the hardware and screen displays. Sitting in close proximity around the screen and touching the screen surface is required when teaching with such tools. These aspects cause significant learning challenges when faced with social distancing restrictions to learning environments. Additionally, as these systems are set up as a single, horizontal terminal, there is limited individual interaction with the system due to the size and orientation of the screen.

Other commercial anatomical 3D visualization products, like the Human Studio from BioDigital [3], offer a freemium-style web-based visualization platform where users can interact with anatomical 3D models. Paid subscriptions to the Human Studio give users access to visualizations of disease and trauma case studies of individual body parts, as well as animated and annotated tours of anatomical structures. BioDigital also provides its platform and custom modeling services to embed 3D models in third-party applications. Similarly, ZygoteBody [17] (previously known as GoogleBody) is another freemium-style product that offers free access to a single 3D anatomical model of the male and female human bodies. However, it offers only the basic features of removing exterior layers to view internal structures. Advanced features such as scene creation, the ability to annotate anatomical parts, or object hierarchies require a paid subscription. The major drawbacks of these platforms are their cost, lack of standardization, and proprietary software sources, hindering their extension by the interested community.

Recently, Santiago G. Izard et al. [10] detailed a proposed system called Nextmed as a multipurpose automatic segmentation, reconstruction, and visualization platform. Directly taking medical scan data such as computed tomography (CT) or magnetic resonance imaging (MRI) scans and creating interactive 3D models using computer vision-based algorithms for 3D segmentation. These models would then become interactive through the use of augmented reality (AR) and/or virtual reality (VR) techniques. The ability to directly convert medical scan data adds value to these systems in clinical and educational settings. However, the 3D display systems

employed are prohibitively expensive and difficult to configure for medical students.

Qi Duan et al. [6] introduced a system named SenseCare, a medical image informatics and 3D visualization platform. The project, however, suggests the use of more advanced artificial intelligence (AI) tool-kits to assist in diagnostic imaging recognition and segmentation when compared with Nextmed. Designed as a web-based platform, SenseCare would focus on medical image processing and AI analysis to act as a research platform for clinical settings. While the use of AI algorithms certainly allows for increased flexibility in analyzing patient medical scans, the complexity and time requirements of training artificial neural networks are not useful for novice medical students and are suitable only for advanced (specialized) training.

Previously, we proposed a simple prototype based on the X3D (ISO) open standard [9] for a web-based simulation environment allowing interaction with 3D anatomical models. The prototype facilitated long-term retention and understanding of neuroanatomy. Our current proposal is expanding the early prototype with a repository of anatomical models, including volumetric medical data. We propose an ecosystem where developers, students, and educators come together to facilitate knowledge transfer in an open, communication-rich environment. The current project builds on the accumulated experience and the feedback over the past decade from medical students and faculty. The new prototype includes a repository of 3D volumetric data visualization capability through 2D section planes, alpha-value transparencies, and 3D polygonal decomposition, built on top of the X3D ISO standard. X3D model developers and medical practitioners may participate by adding their anatomical models to the repository through the web-based user interface.

3 X3D WEB REPOSITORY - WEB ARCHITECTURE

The ability to simulate 3D dissection in a simple web-browser provides an increased learning value as it eliminates the complexity of software tools and the practical risk of damaging tissue structures during an examination. The proposed X3D repository is currently building up as the number of anatomical models available will increase in time driven by the community of users. A critical factor for its success is the openness towards interested community members to participate and add 3D content. Moreover, utilizing a web-based tiered architecture rather than a proprietary software system, our open-source repository has far greater accessibility. The ability to access the repository anytime from any location in a simple Web-browser allows breaks the digital-divide among participants. Model data is decentralized and can be updated from any location rather than being centralized.

The system employs a flexible 3-Tier LAMP architecture to decouple the functionality and improve functional modules' independence. X3D, HTML, CSS, and JavaScript provide the required front-end functionality, following the latest standards in HTML5 to ensure XML compliance and reliability across multiple browsers and devices, including handheld devices. Middle tier operations are handled using PHP, deployed Linux-based Apache Web server. Back-end data management is done through the use of the popular,

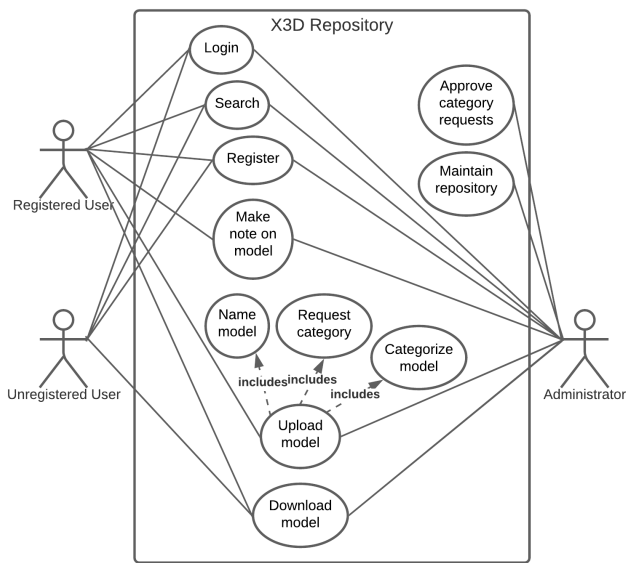


Figure 1: UML Use-case Diagram Showing Possible Interactions with the X3D Visualization System

open-source MySQL. Cloud-based data storage with data replication services is currently being explored to ensure scalability to hundreds of thousands of simultaneous users.

As illustrated in Figure 1, the UML use-case diagram detailing the main actors and their interactions with the system contains: registered users, non-registered users, and administrators. Registered users may explore the repository, as well as add to the repository's X3D anatomical models. Non-registered users can explore the existing 3D models but cannot add models. The administrators will help mitigate potential problems in the system, insert new categories, and mitigate potential user conflicts. The proposed system may evolve into an augmentation for a complex Learning Management System and enrich both the students' experience with 3D content and the instructors' ability to convey medical knowledge.

4 3D MODELS AND INTERACTION FEATURES

The anatomical 3D models can be generated using CT/MRI data, and X3D scripting within the X3DOM [1] and X_ITE software frameworks [5]. The current repository consists of three distinct types of 3D models, sourced from an example archive from Web3D [4] [16]. Interaction with the models is possible for any user with standard input/output methods, either with a mouse, keyboard, and monitor, or touchscreens for mobile devices. Each 3D model page features visual instructions for specific controls, as illustrated in Figure 2. These controls allow users to zoom, rotate, and pan their viewpoint around the model. Users also have access to an options menu built into the X_ITE menu browser platform.

These basic controls are uniform across all the available models, with similar functionality on mobile devices. The web-based interface provides users with a uniform and intuitive experience for interacting with both simple models and more complicated

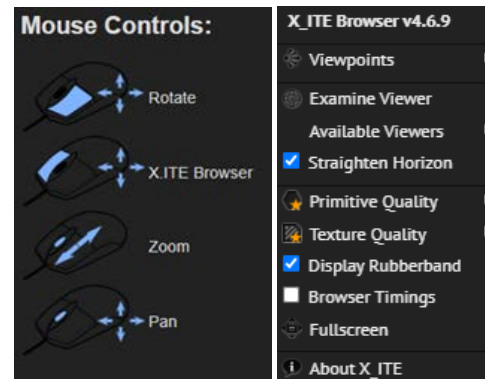


Figure 2: Model Controls: (Left) Mouse Controls; (Right) X_ITE Menu Browser Menu Options

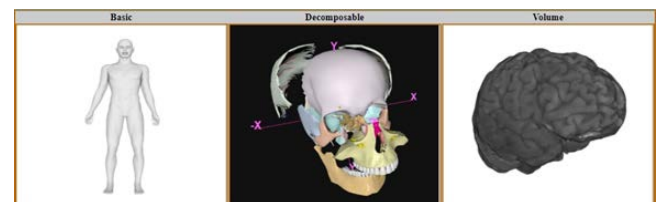


Figure 3: Model Types

structures. The repository has three types of interactive models, as illustrated in Figure 3.

4.1 X3D Static Anatomical Models

The first type of model is a rigid, static 3D model created as a single object which can be manipulated in 3D space but does not contain any volumetric data. These 3D anatomical models are created in X3D by rendering polygonal sets. Content can also be generated from 3D scan files or CT/MRI data. These anatomical models may be used to convey students additional information about singular structures where surface details are most important. High-resolution scans of medical case studies can be used to expose students to a wide variety of reality-based scenarios. An example of static models is illustrated in Figure 4.

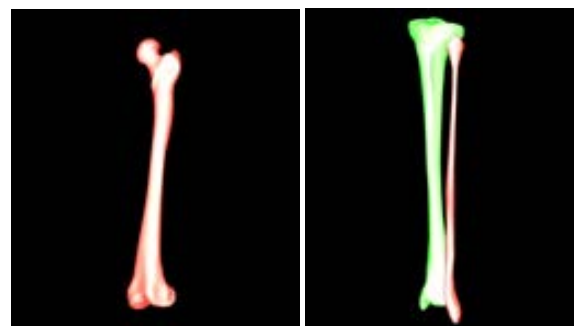


Figure 4: (Left) Left Femur Bone Model, (Right) Left Fibula and Tibia Model

4.2 Decomposable X3D Anatomical Models

Decomposable 3D models are more complex anatomical structures that have various options for decomposition and annotation of individual parts. These are created as a 3D assembly of basic models where each individual structure is a separate node with a particular color and texture property. These nodes can be programmed in X3D to provide animation, emphasizing the anatomical constraints inherent in these structures. Figure 5 provides an illustrative example of a 3D skull - bone structures - and its decomposed/expanded 3D view.

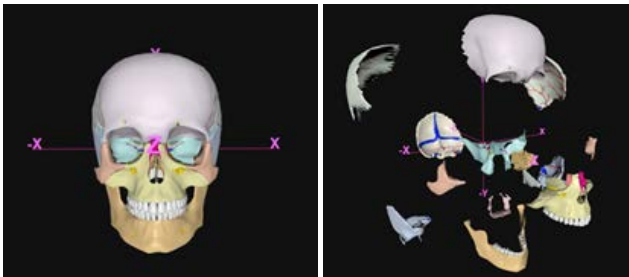


Figure 5: (Left) Decomposable Human Skull Model, (Right) Decomposable Human Skull Exploded View

Such 3D models may facilitate the student's understanding of the physical relationships between different anatomical structures. Complex motions can also be programmed with adequate time and medical guidance to ensure anatomical accuracy, providing users with accurate representations of anatomical movement constraints. Individual nodes can be moved separately or as a whole to give the users a better understanding of the interaction between various bones and tissue structures, enabling users a better understanding of anatomical structures movement in real-world environments, as well as special cases (e.g. accident cases). The decomposable functionality is supported through slider controls in addition to basic controls for the manipulation of individual 3D component parts. It is also possible to disable individual objects to show the structures underneath to exemplify both individual and overall relationships among individual objects, helping students understand and plan complex medical procedures.

4.3 Volumetric Anatomical Data Visualization

The third model type is the volumetric one. The 3D volumes are generated by layering data from CT/MRI scans to create a model with the ability to 'slice' through interior structures using a 2D plane. Beyond the basic control functionality, these models have rendered data throughout their volume. By using the zoom function, users are able to scroll through the model and view deep anatomical structures. An example is presented in Figure 6.

Volumetric models show internal details of structures. They are created by compiling medical imaging scans to form a cohesive object. These can be extremely detailed based on the original scan's resolution, allowing for observation of internal structures that could be damaged during standard dissection procedures. A 2016 study

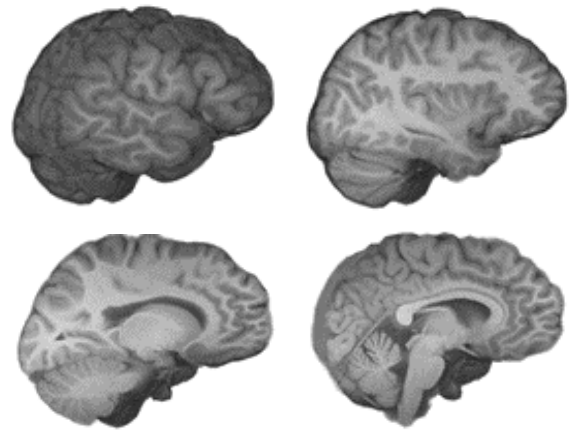


Figure 6: Volumetric Data and Volume Cutting Planes

[14] suggests that the use of 3D renderings can increase an individual's understanding and test scores of students when compared to traditional lecture and gross anatomy laboratory environments.

5 CONCLUSIONS

The realities of Covid-19 have generated physical barriers to learning and created an increased focus on ways to leverage online technology to deliver high-quality remote instruction effectively. Further complicating the situation, many students find themselves and their families in difficult circumstances, particularly financially, and they might not be able to cope with the additional cost and complexity of virtual learning software. As many institutions are moving to online educational technology, specifically in medical education, the access and ability to study 3D anatomical models online in a simple Web-browser can enhance information delivery and support both students and instructors. The proposed repository is web-based, open-source and freely available for use, enhancement and collaboration. X3D is a robust ISO standard for web-based 3D content delivery with many potential applications in the medical domain [8]. An X3D specific expansion into the medical domain, the MedX3D standard has been proposed by the Web3D medical working group [11]. The ability to automate medical CT/MRI scan data into Web3D applications in the near future is very possible and will greatly help expand such 3D repositories. Such systems have the potential to empower educators with the ability to upload scans of novel cases to provide students with real-life simulation scenarios and enhance their training.

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